

PHAGOCYTOSIS

Phagocytosis is the biological process through which cells engulf and digest solid particles or other cells. This mechanism is crucial for both single-celled organisms, such as amoebas, which use it as a means of nourishment, and for multicellular organisms, where it plays a vital role in the immune system. In humans, specialized immune cells known as phagocytes, particularly macrophages and neutrophils, are responsible for this process. Macrophages arise from monocytes and are adept at identifying and ingesting pathogens, aided by proteins called antibodies that mark invaders for destruction. Once a pathogen is engulfed, it is enclosed in a structure called a phagosome, which then merges with a lysosome to form a phagolysosome. This environment becomes acidic, allowing digestive enzymes to break down the foreign material. Neutrophils, the most abundant white blood cells, act quickly to neutralize threats using similar methods and can emit reactive molecules to destroy invaders. This intricate system underscores the essential role of phagocytosis in maintaining health and defending against infections.

Phagocytosis is the process by which a cell engulfs and consumes other cells or solid particles. In single-celled organisms such as the amoeba, phagocytosis is a method of feeding. In higher organisms, the process is a vital part of the immune system and is carried out by specialized cells called macrophages and certain white blood cells. These cells, known as phagocytes, absorb and digest invading organisms such as bacteria, viruses, or other foreign bodies. The term phagocytosis comes from the Greek words *phagein*, meaning "to eat," and *cytos*, meaning "cell." Phagocytosis is a type of endocytosis in which a cell ingests solid particles. Pinocytosis, or cellular drinking, occurs when the cell consumes liquids.

Microscopic organisms that consist of a single cell sustain themselves through the process of phagotrophic nutrition. When an amoeba, for example, detects the presence of a bacterium, it crawls toward its target. When it makes contact, the amoeba sends out "arms" called pseudopodia formed from its cell membrane. Some single-celled organisms use hair-like structures called cilia to attach themselves to their prey. The pseudopodia surround the bacterium and encase it in an internal cavity called a vesicle. Before the bacterium can reproduce, the amoeba releases other vesicles packed with digestive enzymes, proteins that produce a chemical reaction in organic substances. These vesicles attach themselves to the food, break down the bacterium, and absorb the nutrients.

In the human body, phagocytosis occurs in many cells, but the most common are white blood cells found in the immune system. These cells make up about 1 percent of the body's blood and act as natural infection fighters. Of the five types of white blood cells, monocytes and neutrophils are the most accomplished phagocytes. Monocytes are the largest white blood cells and are created in the bone marrow, a soft substance in the center of the bones. After monocytes are released into the bloodstream, they travel to the site of an infection and enter the body's tissue. They then develop into specialized infection-devouring cells called macrophages. The large size of the cells makes it easier for them to consume foreign invaders.

Since macrophages must come into contact with an infectious organism before consuming it, the body must first find a way to let the macrophage know which object to target. One way is for other white blood cells to produce proteins called antibodies that attach themselves to certain molecular patterns on the surface of the invading cell. These antibodies act as a homing beacon for the macrophages, marking a particular cell for destruction. Some macrophages have specialized receptors on their surface designed to search for specific antibodies and bind with them. Others have receptors designed to seek out and bind with certain molecules produced by the bacteria itself.

When the receptors on the surface of the macrophage bind with the target, the macrophage begins drawing the cell inward, encircling it, and closing around it. The target becomes enclosed inside an internal bubble called a phagosome. This structure then fuses with an enzyme-filled bubble inside the cell called a lysosome. Once formed, the merged structure, known as a phagolysosome, reduces its internal pH level. A pH level is a measure of acidity; the lower the pH level, the more acidic the environment. This kills or neutralizes the target before it has a chance to reproduce, allowing the macrophage's enzymes to break it down for absorption. The resulting waste product is later expelled from the macrophage.

Some viruses have the ability to trick a macrophage and hijack it to replicate inside the cell. The influenza virus, for example, uses the drop in pH levels to escape from the phagolysosome and gain access to the macrophage's interior. When a macrophage becomes infected by a virus, the body's other phagocytes will then mark it as a target.

Neutrophils are the most numerous white blood cells, making up about 60 percent of the body's total white blood cells. The first line of defense to arrive at the site of an infection, neutrophils act as natural assassins, using several methods to kill their targets. They are also produced in the bone marrow and live for only a few hours. Their primary method of attack is similar to the macrophage; they engulf an invader and digest it through enzymes released from granules, or grain-like shapes, within their structures. Neutrophils can also emit a burst of highly reactive oxygen molecules that can damage and destroy invading cells. Neutrophils and other phagocytosis can also kill their engulfed prey by releasing antimicrobial proteins specifically designed to destroy bacteria or producing binding proteins that interfere with a bacterium's ability to reproduce.